An Empirical Study of Certified Feature Selection for Robust Hypothesis Testing

7 October 2026

Abstract

We evaluate global feature selection for a two-class Wasserstein robust testing criterion using four synthetic alternatives and two public datasets. Independent selection, classifier-fitting and test samples separate optimization accuracy from statistical utility. Across 159 solved small gate/enumeration pairs, the largest recorded objective difference is zero, with 0 incompatible bound intervals. On mean-shift data, 30/30 global runs meet the optimization tolerance, but common-test power is 0.928 versus 0.992 for marginal screening. The risk-scaled gate yields stronger lower models without consistently reducing runtime. Empirical classifier certificates do not establish population coverage, and an exploratory full-fraction master does not improve elapsed time on its paired instances. The records support the finite optimization construction while exposing limitations of computational efficiency and small-sample statistical selection.

1 Introduction

Selecting a small set of measurements for hypothesis testing requires both a selection criterion and a tractable search over feature subsets. We study a criterion based on two class-conditional Wasserstein uncertainty sets and a global-cut algorithm derived from the logical-constraint framework of Bertsimas et al. [1]. The accompanying theory note establishes integer exactness and valid optimization bounds. Here we evaluate computation and independent statistical performance.

The study adapts the independent testing and recovery protocol of Wang et al. [4], the small-sample robust-testing setting of Xie et al. [2, 3], and separate downstream fitting as in Swaroop et al. [6]. Six selectors share the same data and evaluation models. We report both favorable and unfavorable outcomes and distinguish minimization of the empirical robust criterion from usefulness on an unknown population.

2 Method summary

Let $\widehat{P}_k$ be the empirical distribution of class $k \in \{0,1\}$ and $\mathcal{U}_k = \{P : W_1(P, \widehat{P}_k) \le \theta_k\}$, with positive radii and standardized L1 transport cost. A mask z selects at most d of the D coordinates, and the test score g depends only on $x_{S(z)}$. The target is

$$\mathrm{OPT} = \min_{\substack{z\in\{0,1\}^D \\ \mathbf{1}^\top z \le d}} V(z), \qquad V(z) = \inf_{g=g(x_{S(z)})} \left\{ \sup_{P_0\in\mathcal{U}_0} \mathbb{E}_{P_0}\Phi(-g) + \sup_{P_1\in\mathcal{U}_1} \mathbb{E}_{P_1}\Phi(g) \right\}, \tag{1}$$

where $\Phi(v) = (1+v)_+/2$. A positive score favors class zero. Clipping scores to $[-1,1]$ gives a randomized probability $\pi_0 = (1+g)/2$; consequently V is a sum of classwise errors and $V/2$ is its balanced-average counterpart. The constant test has risk $B = 1$. A test may ignore extra coordinates, so the search can use the equality $\mathbf{1}^\top z = d$.

For each fixed support, an exact finite linear program returns a feasible whole-space test and a dual lower certificate. Its coordinate prices satisfy $0 \le w_{kj} \le \lambda_k$, where λ_k prices class-k transport. We compare three binary-equivalent gates:

$$\begin{aligned} &\text{Box:} && w_{kj} \le (B/\theta_k) z_j, \\ &\text{Triangle:} && \theta_0 w_{0j} + \theta_1 w_{1j} \le B z_j, \\ &\text{Risk-scaled:} && \theta_0 w_{0j} + \theta_1 w_{1j} \le \eta z_j. \end{aligned} \tag{2}$$

Here η bounds the test objective. The box constants bound multipliers on the constant-test sublevel set; they are not a population-radius assumption. The risk-scaled dual supplies a convex fraction and hence

a global tangent. The integer master minimizes the accumulated lower model over all budget-feasible supports. Each new support triggers a complete oracle refit; the best feasible original test supplies the upper bound $\bar{v}$, while the master supplies a global lower bound $\underline{v}$. Valid bounds satisfy $\underline{v} \leq \mathrm{OPT} \leq \bar{v}$. A run is declared numerically optimal only when

$$\bar{v} - \underline{v} \leq 10^{-6} + 10^{-5}|\bar{v}|. \tag{3}$$

The primary risk method adds five fractional root rounds. A no-root variant isolates their computational effect. All variants start from the same marginal MMD support. The companion note gives the finite reduction, deployment construction, dual derivations and proofs; Appendix D describes the exploratory full-fraction master.

3 Experimental design

3.1 Data and evaluation splits

The synthetic study uses $D = 12$, a three-coordinate signal set and budget $d = 3$. Class zero is standard Gaussian; class one changes only the signal block as specified in Table 1. Remaining coordinates are independent standard Gaussian noise. Coordinate positions are shuffled for each seed, and recovery uses their underlying semantic identities. The dependence alternative has identical marginal distributions, so

Table 1: Synthetic alternatives before standardization.

Mechanism	Class-one signal block
Mean	Shift each coordinate by 0.9.
Variance	Increase each coordinate's standard deviation to 1.8.
Dependence	Standard-normal marginals with equicorrelation 0.65.
Mixture	Add one shared random sign times 1.5 to all three coordinates.

useful signal requires joint measurements. The mixture preserves equal means while changing marginal shape and joint structure.

Each of 30 training seeds provides disjoint selection and downstream-training samples of 20 observations per class, and 1,000 fresh classification observations per class. Centering and scaling are fitted on pooled selection data and frozen for every selector and later split. The primary radii are $\theta_0 = \theta_1 = 0.25$. Equality testing uses twenty fresh alternative batches and twenty null batches per seed, each with 40 observations per class; null groups are both drawn from class zero.

The real-data study uses UCI WDBC (569 records, 30 features) and Ionosphere (351 records, 34 features), excluding identifiers and labels [8, 9]. Twenty stratified splits allocate 20 observations per class to selection and a disjoint 20 to downstream fitting; remaining observations are held out. The budget and radii match the synthetic study. Source hashes, class mappings and complete settings are retained in Appendix A.

3.2 Selectors and reproduction scope

The six budget-matched selectors are global robust-risk selection, fully refitted robust greedy, marginal MMD screening, Wang additive-MMD with the paper objective, L1 logistic ranking and seeded random selection. Robust greedy minimizes the same criterion at each forward step; marginal screening ranks one-dimensional unbiased RBF MMD. Logistic ranking uses absolute coefficients from an L1 fit with liblinear and $C = 1$. All methods acquire three features. Known-signal and all-feature supports are separate diagnostics.

The Wang baseline optimizes an additive coordinate kernel under $\|\omega\|_2 = 1$ and $\|\omega\|_0 \leq d$, using the paper's mean-minus-variance criterion [4]. Fixed-support weights are globally fitted; supports are

enumerated for $D \leq 12$ and searched approximately at larger D. The inspected author revision instead maximizes a positive variance term. We therefore reproduce the paper objective, not that revision's numerical pipeline. Appendix B records the coefficients and provenance.

The primary testing evaluation applies the same joint RBF statistic to each selected support. Wang's separately reported native signed additive statistic retains its learned weights; its power is not interchangeable with common support-only power. Both tests freeze parameters before final labels are used. The multi-population objective of Swaroop et al. and the safe union-screening task of Hanada et al. [7] do not define equivalent fixed-budget comparators for these two class-conditional balls.

3.3 Metrics and uncertainty

Optimization outcomes include solved fraction, charged time, bounds, absolute gap, oracle/master time and distinct integer supports. Runs share a 30-second selection allowance and use (3); all failures and time limits remain in their denominators. The original-run and retry costs are charged together.

Equality-test power and null rejection use the unbiased joint RBF MMD with a selection-fitted median bandwidth. With 199 random label permutations, the plus-one p-value is $(1 + \#\{T_{\text{perm}} \geq T_{\text{obs}}\})/200$, and the level is 0.05. Signal recall is $|S \cap T_*|/|T_*|$. Under the equal support and signal sizes here, precision and F1 equal recall, while FDP and NDP both equal one minus recall; they are not independent endpoints.

Each support is independently fitted with the same robust envelope test and RBF SVM ($C = 1$; downstream median bandwidth). Prediction is measured by thresholded balanced error, randomized balanced error, AUC and predictive cross-entropy. Randomized balanced error is $\frac{1}{2}[\overline{1 - \pi_0(X)} + \overline{\pi_0(Y)}]$; the empirical half-hinge average is identical. Hard decisions use threshold 0.5, with ties assigned to class zero.

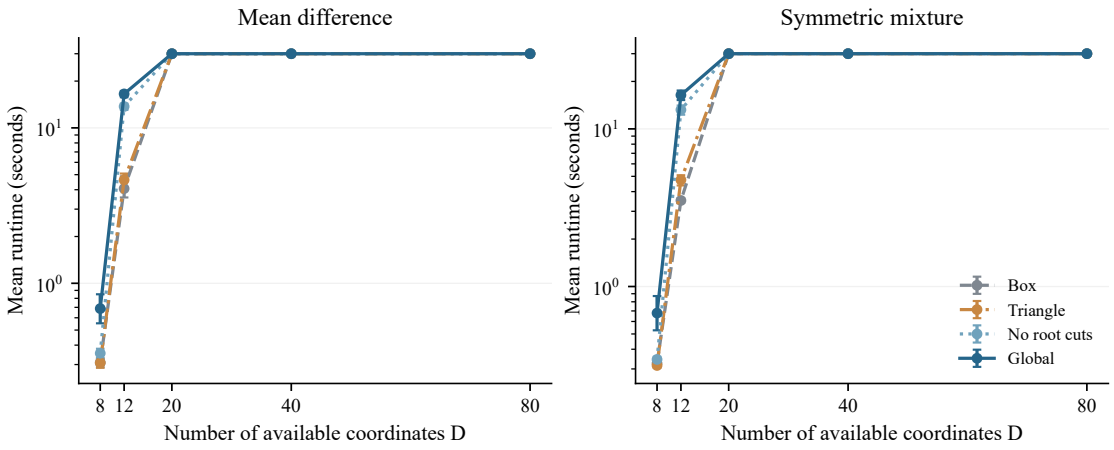
Synthetic intervals are 95% percentile bootstrap intervals from 2,000 resamples of training seeds. Conditional batches are averaged within seed first, and method differences retain pairing. Real results are mean $\pm$ repeated-split SD, because records are reused across splits. Complete metric tables and effective counts appear in Appendix C.

4 Computational results

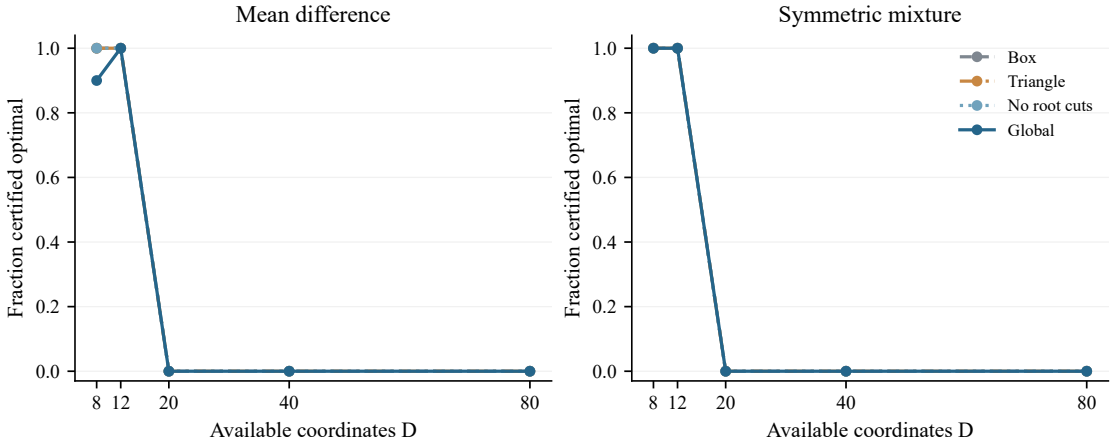
Small comparisons use mean and mixture data at $(D, d) = (8, 2), (12, 3)$, eight source atoms per class and ten paired seeds. Scaling uses $D = 20, 40, 80$, budget three, ten atoms per class and five seeds. Box, triangle, risk without roots and risk with roots share the objective, warm support and numerical controls; small cases also use enumeration. The 159 solved gate/reference pairs have maximum absolute objective difference zero; 0 recorded intervals are incompatible with enumeration at the combined numerical tolerances.

Table 2: Gate comparison at $D = 12, d = 3$. Charged time is mean seconds [95% seed interval]; solved counts retain all ten outcomes. Full dimensions, gaps and component times are in Appendix C.

Method	Mean solved	Mean time (s)	Mixture solved	Mixture time (s)
Box	10/10	4.07 [3.57, 4.88]	10/10	3.52 [3.40, 3.63]
Triangle	10/10	4.63 [4.16, 5.08]	10/10	4.75 [4.38, 5.08]
Risk, no root	10/10	13.73 [13.05, 14.32]	10/10	13.22 [12.30, 14.07]
Global risk	10/10	16.54 [15.74, 17.28]	10/10	16.39 [15.24, 17.52]
Enumeration	10/10	1.38 [1.34, 1.40]	10/10	1.37 [1.35, 1.39]



(a) Charged selection time; 95% seed-bootstrap intervals.



(b) Numerically solved fraction; time limits and failures remain in the denominator.

Figure 1: Primary gate comparison under the common selection allowance. All recorded outcomes are retained. Root and master work and numerical retries contribute to charged time.

At $D = 12$ for mean data, the risk method visits 197.8 supports on average versus 220.0 for box, but spends 15.16 s in the master versus 2.51 s. At $D = 12$ for mixture data, the risk method visits 197.0 supports on average versus 220.0 for box, but spends 15.02 s in the master versus 2.11 s. At $D \geq 20$, 120/120 gate runs end at the selection limit. The achieved gaps remain informative even though no exact solution is recorded. At $D = 20$ for mean data, mean gap is 0.167 for risk and 0.276 for box. At $D = 20$ for mixture data, mean gap is 0.151 for risk and 0.263 for box.

5 Synthetic statistical results

5.1 Testing and recovery

Common-test null rejection means range from 0.040 to 0.060 at nominal level 0.05. Figures 2 and 3 show the mechanism-specific power and recovery results.

For mean data, global selection has common-test power 0.928 versus 0.992 for marginal screening. The paired difference is -6.3 percentage points (95% CI [-14.3, -1.3]; $n = 30$). Signal recall is 0.533 versus 0.756. All 30/30 global runs meet the optimization tolerance, but greedy power is 0.965.

For variance data, global selection has common-test power 0.360 versus 0.405 for marginal screening. The paired difference is -4.5 percentage points (95% CI [-19.7, +11.0]; $n = 30$). Signal recall is 0.433

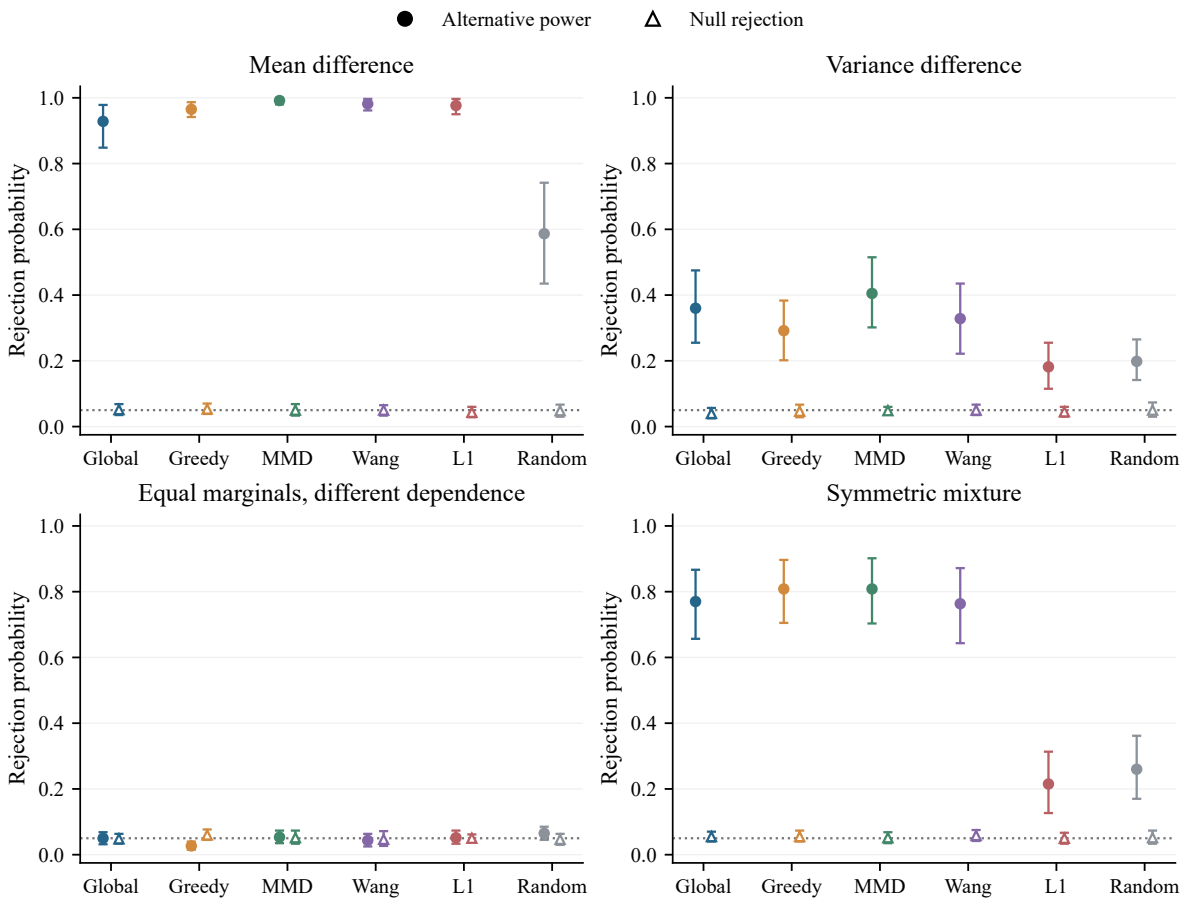


Figure 2: Common-support two-sample power and null rejection at level 0.05. Points average twenty batches within each of thirty training seeds; intervals resample those seeds.

versus 0.533. Wang's native signed additive statistic has power 0.518 and null rejection 0.052, compared with common support-only power 0.328 ($n = 30$).

For dependence data, global selection has common-test power 0.050 versus 0.053 for marginal screening. The paired difference is -0.3 percentage points (95% CI [-2.8, +2.0]; $n = 30$). Signal recall is 0.256 versus 0.267.

For mixture data, global selection has common-test power 0.770 versus 0.808 for marginal screening. The paired difference is -3.8 percentage points (95% CI [-20.3, +11.7]; $n = 30$). Signal recall is 0.633 versus 0.711. Wang's native signed additive statistic has power 0.945 and null rejection 0.040, compared with common support-only power 0.763 ($n = 30$).

Independent classification follows a similar pattern on mean data: robust-classifier balanced error is 0.382 for global selection and 0.346 for screening; SVM error is 0.315 and 0.279, respectively.

For the separately fitted robust classifier, 120/120 independent randomized errors exceed their own empirical head bound divided by two (Table 3). The corresponding 120 valid source-atom checks have 0 violations above 10^{-7}. This empirical-ball certificate is distinct from the selection objective. The evaluated population was not established to lie in that ball; the held-out excess is therefore not a certificate failure or a population-coverage result.

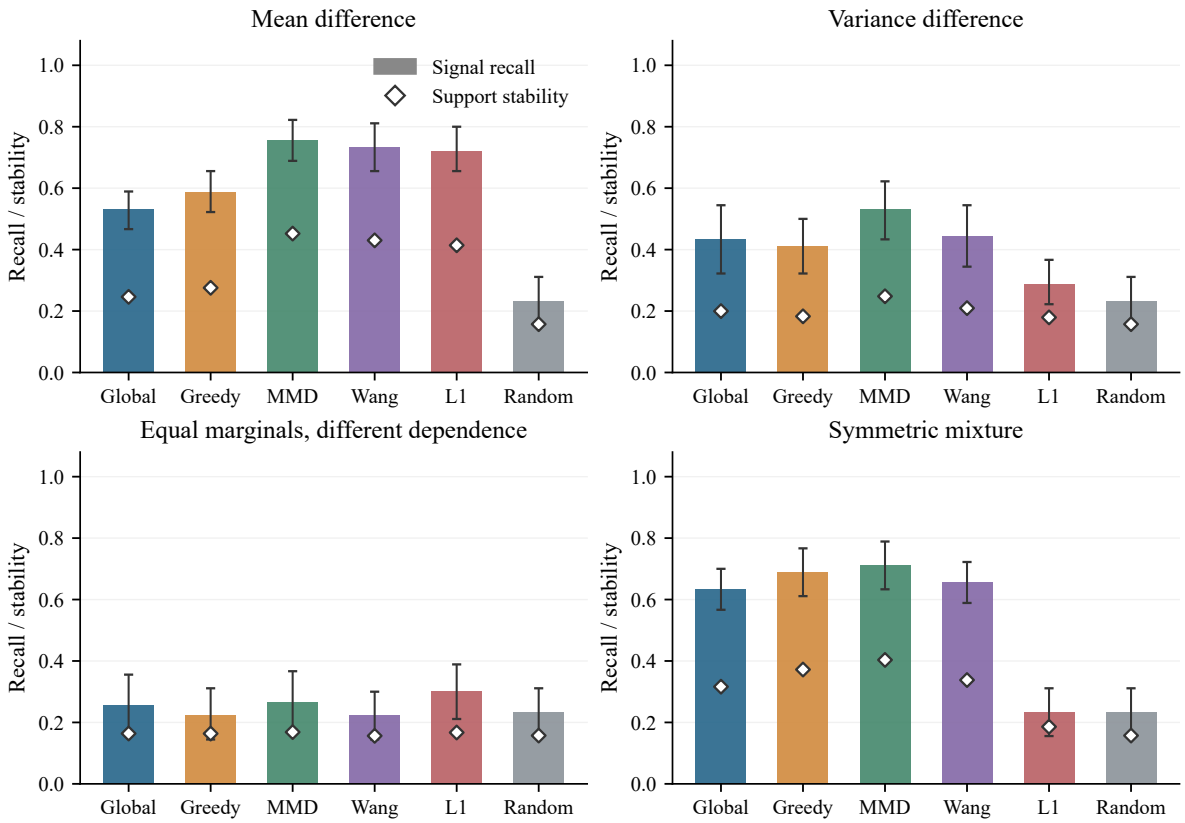


Figure 3: Signal recall and support stability. Bars show recall with 95% seed-bootstrap intervals; diamonds show descriptive pairwise Jaccard stability.

Table 3: Refitted empirical head risk and independent randomized error for global selection. Means use paired training seeds. Above-bound counts compare each seed at tolerance 10^{-8}; the final column is the maximum budget-feasible source-atom violation.

Mechanism	n	Head risk / 2	Test error	Above bound	Source violation
Mean difference	30	0.251	0.448	30/30	0.000
Variance difference	30	0.275	0.482	30/30	0.000
Equal marginals, different dependence	30	0.287	0.497	30/30	0.000
Symmetric mixture	30	0.248	0.423	30/30	0.000

5.2 Prediction and radius sensitivity

The common robust and SVM classifiers assess support utility on independent individual observations. Full classifier, ranking and probability-loss results are in Appendix C. We also translate held-out class observations toward the opposite class along one common downstream-mean direction of L1 norm one, using magnitudes $0, 0.25, 0.50$. The worst measured error is a maximum over these three shifts, rather than a Wasserstein supremum. A separate source-atom translation has an explicit budget-feasible coupling and checks the fitted certificate directly.

The predeclared radius analysis refits the procedure at $0.10, 0.25, 0.50$ for mean and mixture data on ten seeds. Both selection and downstream robust fitting change with radius; this comparison does not isolate selection alone. For mean data, from radius 0.10 to 0.50, power changes from 0.790 to 0.955, randomized balanced error from 0.494 to 0.426, and worst measured thresholded stress error from 0.488

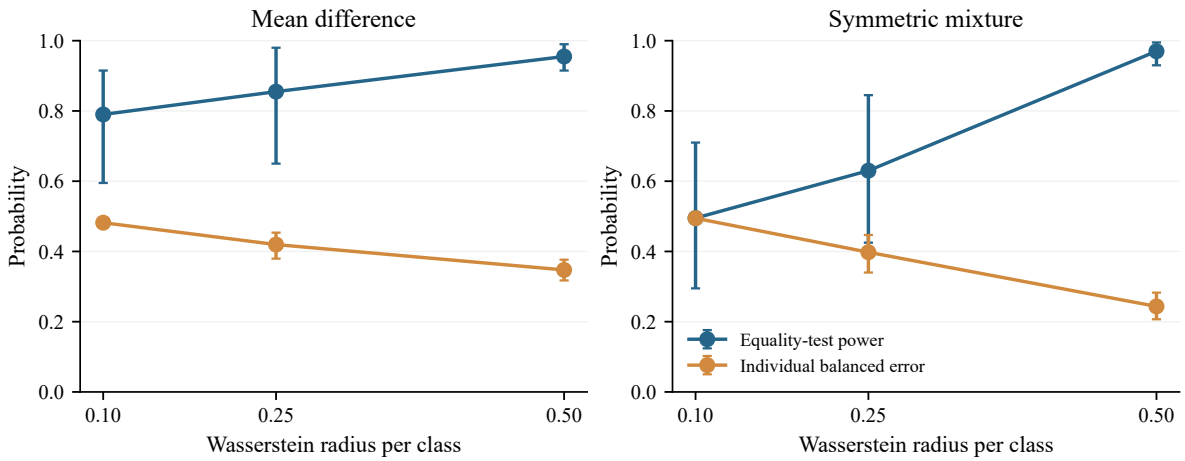


Figure 4: Sensitivity to the three preset radii. Each setting uses the same seed-level data and independent evaluation protocol; intervals resample ten training seeds. The primary radius remains 0.25.

to 0.380. The observed power averages increase across these settings. The support and robust classifier are both refitted, so these changes describe their joint sensitivity.

For mixture data, from radius 0.10 to 0.50, power changes from 0.495 to 0.970, randomized balanced error from 0.486 to 0.371, and worst measured thresholded stress error from 0.499 to 0.256. The observed power averages increase across these settings. The support and robust classifier are both refitted, so these changes describe their joint sensitivity.

6 Real-data results

Both datasets use the common budget and independently fitted classifiers. Global searches are evaluated through their retained supports and recorded statuses. Table 4 reports predictive variability over the twenty predefined splits. Selection gaps, stress results and feature frequencies are provided in Appendix C. Global searches reach the selection limit in 40/40 splits. Their classifier results use the retained incumbent supports. On WDBC, global selection has higher observed robust classifier error and lower observed SVM error than marginal screening. On Ionosphere, global selection has higher observed robust classifier error and higher observed SVM error than marginal screening.

Table 4: Common classifiers on real datasets. Entries are mean ± repeated-split SD. BER is thresholded balanced error; AUC uses the robust classifier's scores.

Dataset	Selector	n	Robust BER	SVM BER	Robust AUC
WDBC	Global risk	20	0.112±0.031	0.081±0.017	0.965±0.020
	Robust greedy	20	0.099±0.021	0.080±0.021	0.973±0.013
	MMD screening	20	0.091±0.021	0.084±0.017	0.973±0.011
	Additive MMD	20	0.090±0.019	0.082±0.014	0.975±0.010
	L1 logistic	20	0.097±0.035	0.078±0.020	0.971±0.017
	Random	20	0.192±0.089	0.156±0.078	0.893±0.093
Ionosphere	Global risk	20	0.243±0.059	0.251±0.094	0.869±0.053
	Robust greedy	20	0.217±0.052	0.202±0.081	0.885±0.045
	MMD screening	20	0.223±0.055	0.195±0.053	0.865±0.055
	Additive MMD	20	0.209±0.047	0.181±0.058	0.877±0.051
	L1 logistic	20	0.227±0.050	0.198±0.049	0.880±0.046
	Random	20	0.270±0.047	0.303±0.069	0.835±0.034

7 Discussion

The experiments support the optimization formulation but do not establish a general statistical advantage of the robust criterion. Solved small instances agree with enumeration, whereas stronger fractional gates did not reduce elapsed time in this master implementation. Independent testing and recovery can remain worse even when the empirical risk is optimized to tolerance. Radius coverage and finite-sample objective choice therefore require further study.

The empirical head bound applies to its specified Wasserstein balls. Independently evaluated error can exceed it when population coverage is unestablished; the relevant comparison is reported separately from the direct source feasibility check. The native Wang statistic similarly distinguishes an end-to-end testing procedure from utility of the same selected support under a common evaluator.

A separately declared full-fraction master reduced support visits on some instances without improving elapsed time overall. Appendix D reports its paired results and exact binary lift. These exploratory results motivate improving master solution rather than treating stronger integer constraints as a runtime guarantee.

The study uses small training samples, moderate computational dimensions, fixed radii and fixed baseline hyperparameters. Repeated real-data splits describe variability within two datasets, not performance across independent populations. No formal multiple-comparison significance claims or clinical claims are made. All recorded outcomes, original numerical failures and replay costs are preserved.

A Protocol and reproducibility

Table 5: Frozen settings. The protocol date is 3 October 2026.

Experiment	Settings
Statistical	Four alternatives; $D = 12$, signal and budget three; seeds 0–29; selection and downstream training 20 per class; classification 1,000 per class; twenty alternative and twenty null batches per seed, each 40 per class; 199 permutations; radius 0.25.
Small computation	Mean and mixture; $(D, d) = (8, 2), (12, 3)$; eight atoms per class; seeds 0–9; four gate/root variants and enumeration.
Scaling	Mean and mixture; $D = 20, 40, 80$, $d = 3$; ten atoms per class; seeds 0–4; four gate/root variants.
Real data	WDBC and Ionosphere; seeds 0–19; disjoint selection/downstream samples of 20 per class, remaining records held out; budget three, radius 0.25.
Radius	Mean and mixture; seeds 0–9; statistical sample sizes; global risk selection at $0.10, 0.25, 0.50$.
Full-fraction master	Mean and mixture; eight atoms per class; seeds 10–14; $(D, d) = (8, 2), (12, 3), (20, 3)$; tangent versus ratio-plus-tangent; five root rounds, no core-point selection.
Controls	Thirty-second selection allowance, shared warm support, 10^{-6} absolute and 10^{-5} relative tolerance; one solver/BLAS thread; paired data, scaling and evaluation.

A.1 Numerical failures and replay

The frozen original failure inventory contains 12 runs. The 12 replays ended in 1 master failure, 5 numerical optima, and 6 terminations at the time limit. The final collection has 3 failures, including 2 outside the original replay cohort. Final diagnostics record 66 identical-model master retries.

Archived failed attempts charged 112.47 selection seconds and 112.87 end-to-end seconds. The final record timings include those original costs in addition to rerun costs. The rerun receives only the original 30-second selection allowance minus the original charged selection time; both attempts share that allowance, subject to modest call/finalization overhead. The appendix lists the retained original and final statuses rather than suppressing the first outcomes. Only status-4 solver errors trigger one same-model retry with presolve disabled. An error return contributes no optimization bound. Original and replayed oracle/master costs are cumulative; replay receives only the unused original selection allowance. The independent numerical probe is not a performance observation.

Table 6: Original replay cohort and final outcomes. Charged time includes the first attempt and replay.

Trial	Original status	Final status	Original (s)	Charged (s)
computational, mean $D = 8$, seed 3; Global risk	master failure	optimal	0.62	1.31
computational, mean $D = 8$, seed 9; Global risk	master failure	master failure	0.44	0.97
computational, mean $D = 12$, seed 3; Box	master failure	optimal	3.94	7.50
computational, mixture $D = 8$, seed 8; Global risk	master failure	optimal	0.66	1.33

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Trial	Original status	Final status	Original (s)	Charged (s)
computational, mixture				
$D = 8$, seed 9; Global risk statistical, mean	master failure	optimal	0.54	1.14
$D = 12$, seed 3; Global risk statistical, mixture	master failure	optimal	1.25	23.36
$D = 12$, seed 1; Global risk statistical, mixture	master failure	time limit	20.27	30.00
$D = 12$, seed 6; Global risk statistical, mixture	master failure	time limit	20.92	30.00
$D = 12$, seed 13; Global risk statistical, variance	master failure	time limit	13.85	30.00
$D = 12$, seed 4; Global risk statistical, variance	master failure	time limit	19.54	30.00
$D = 12$, seed 8; Global risk statistical, variance	master failure	time limit	19.92	30.00
$D = 12$, seed 21; Global risk	master failure	time limit	10.51	30.00

A.2 Verification and provenance

Independent small-instance checks compare the original fixed-support primal with a source-marginal total-transport dual, enumerate all supports, evaluate cuts away from their generating support and check whole-space deployment. They include unequal weights and radii, duplicate observations, empty/full budgets, analytic curves and smaller logistic-cone cases. Additional checks cover splitting, kernel coefficients, global weight fitting, permutation arithmetic, risk normalization and interrupted master bounds.

Each stored run contains its configuration hash, preprocessing, selected observed and semantic coordinates, splits, bounds, status, timing, fitted head and test replicates; traces are separate. The final generator checks all planned keys and compares raw metrics against the public CSV before authoring. The report manifest hashes its inputs and figure assets. Canonical complete results remain in `per_trial.csv` and `summary.json`. Build revision and the original study's verification/software metadata are reported with the generated appendix.

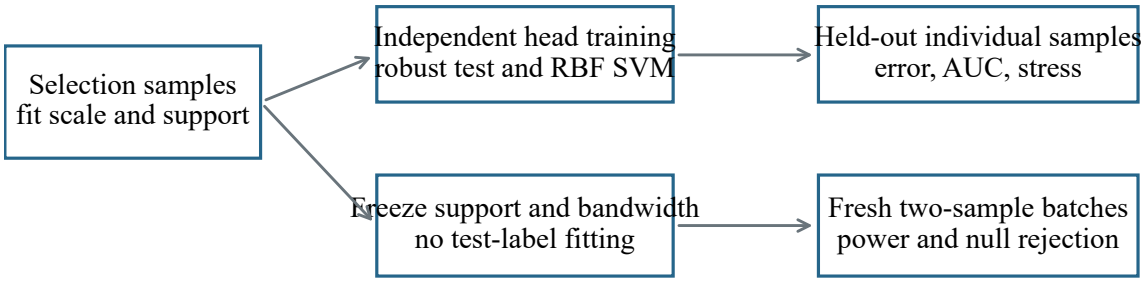


Figure 5: Sample roles in a synthetic training seed. Transformations and support use selection data; downstream fitting and final assessment use separate samples.

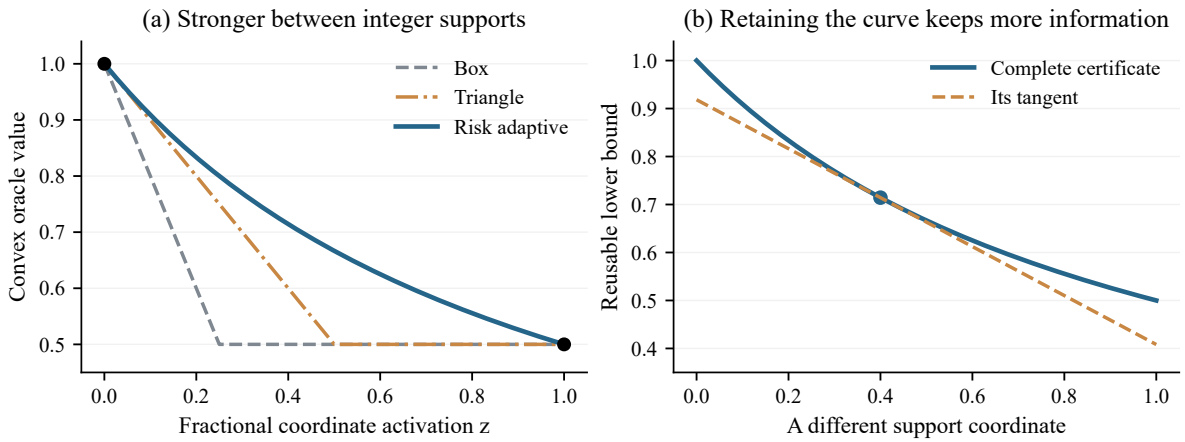


Figure 6: Analytic illustration of gate values and the complete dual fraction. Binary risk is unchanged; fractional lower models differ. Detailed proofs are in the companion theory note.

B Baseline fidelity

For Wang's paired construction, let $h_{il,j} = k_j(x_i, x_l) + k_j(y_i, y_l) - k_j(x_i, y_l) - k_j(y_i, x_l)$. The reproduced mean and row-based variance coefficients are

$$t = \frac{1}{n(n-1)} \sum_{i \neq l} h_{il}, \qquad r_i = \frac{1}{n} \sum_{l} h_{il}, \qquad \hat{\Sigma} = \frac{4}{n} \sum_{i} (r_i - \bar{r})(r_i - \bar{r})^\top, \tag{4}$$

and the criterion is

$$\max_{\|\omega\|_2 = 1,\ \|\omega\|_0 \leq d} t^\top \omega - 0.1\, \omega^\top \hat{\Sigma} \omega. \tag{5}$$

On each support an eigendecomposition and secular norm equation give the global sphere-constrained weight solution, including its hard case. Coordinate bandwidths follow the inspected revision convention: median squared cross-class selection distances, factor one, and a constant-column floor. At larger dimensions the documented forward/swap search is approximate.

The inspected author code [5] maximizes a positive quadratic variance coefficient, rather than the negative coefficient in (5). It also recomputes a cross-class bandwidth using final labels. Our independently written baseline follows the paper objective and freezes bandwidths before final testing. Signed weights are retained; conditional permutation calibration does not require the weighted statistic to be a positive-definite-kernel MMD. This is a criterion reproduction with adapted independent evaluation, not a reproduction of the source's tables. The inspected commit is `031f740e9ca6672751145b16de62cfaf731e98de`. The detailed coefficient fixtures and adaptations are in `docs/baseline_reproduction.md`.

The local source versions are Wang v4 (2024), Xie v1 (2021), Gao v1 (2018), Swaroop v1 (2025) and Bertsimas v3. Unlicensed author source is not redistributed. Dataset URLs, DOIs, CC BY 4.0 licenses, hashes, dimensions, label mappings and excluded columns are in `docs/data_sources.json`.

C Supplementary results

Synthetic entries are mean [95% training-seed bootstrap interval]; real entries are mean ± split SD. Dashes denote absent or inapplicable values. Counts and statuses retain unsuccessful outcomes. Redundant half-hinge and randomized-error aliases are not treated as separate findings.

C.1 Original study metadata

Table 7: Recorded outcome inventory. Complete denotes completed evaluation, not certified optimal selection.

Suite	Planned	Present	Evaluated
statistical	960	960	960
computational	320	320	320
real	240	240	240
sensitivity	60	60	60
ratio	60	60	60

The original study verification reports 106 passed tests in 8.72 seconds. The host is AMD Ryzen 9 7945HX with Radeon Graphics. Summed charged selection time is 2.94 hours; summed end-to-end trial time, including evaluation, is 2.96 hours. These sums include unsuccessful outcomes and recorded remediation costs; they are not elapsed calendar time for concurrent runners.

The LP and MILP engine is SciPy-backed HiGHS 1.12.0 (SciPy 1.18.1). Logistic correctness checks use CVXPY 1.9.3 with Clarabel 0.11.1.

Build revision: a2f76388d42b. Input and configuration SHA-256 hashes are saved with the generated report manifest.

The completed small-instance comparisons contain 159 paired, numerically solved gate/reference outcomes. Their largest absolute feasible-objective difference is 0.000; 0 recorded gate intervals fail the paired enumeration compatibility check at the combined declared tolerances. Agreement validates the shared integer target, rather than a statistical performance gain.

C.2 Full optimization comparisons

Table 8: Mean difference: primary gate comparison. Time is charged mean seconds [95% seed-bootstrap interval], with all outcomes retained. Gap is mean absolute upper-minus-lower bound; visits count distinct integer supports. O: numerical optimum; T: time limit; M: master failure; F: oracle failure; A: accuracy limit.

(D, d)	Method	Solved / n	Time (s)	Gap	Visits	Status
8, 2	Box	10/10	0.31 [0.29, 0.33]	4.0×10^{-7}	28.0	10O
8, 2	Triangle	10/10	0.31 [0.29, 0.33]	6.0×10^{-7}	28.0	10O
8, 2	Risk, no root	10/10	0.35 [0.33, 0.38]	4.0×10^{-7}	28.0	10O
8, 2	Global risk	9/10	0.69 [0.55, 0.85]	0.011	27.3	1M, 9O
8, 2	Enumeration	10/10	0.16 [0.16, 0.17]	1.4×10^{-9}	28.0	10O
12, 3	Box	10/10	4.07 [3.57, 4.88]	1.0×10^{-7}	220.0	10O
12, 3	Triangle	10/10	4.63 [4.16, 5.08]	1.3×10^{-9}	219.9	10O
12, 3	Risk, no root	10/10	13.73 [13.05, 14.32]	4.0×10^{-7}	203.2	10O
12, 3	Global risk	10/10	16.54 [15.74, 17.28]	2.0×10^{-7}	197.8	10O
12, 3	Enumeration	10/10	1.38 [1.34, 1.40]	1.3×10^{-9}	220.0	10O
20, 3	Box	0/5	30.01 [30.00, 30.02]	0.276	808.2	5T
20, 3	Triangle	0/5	30.02 [30.01, 30.03]	0.279	812.2	5T
20, 3	Risk, no root	0/5	30.00 [30.00, 30.00]	0.252	306.4	5T

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(D, d)	Method	Solved / n	Time (s)	Gap	Visits	Status
20, 3	Global risk	0/5	30.00 [30.00, 30.00]	0.167	178.8	5T
40, 3	Box	0/5	30.02 [30.02, 30.03]	0.287	653.0	5T
40, 3	Triangle	0/5	30.01 [30.01, 30.02]	0.287	693.6	5T
40, 3	Risk, no root	0/5	30.04 [30.02, 30.06]	0.292	584.0	5T
40, 3	Global risk	0/5	30.00 [30.00, 30.01]	0.199	122.8	5T
80, 3	Box	0/5	30.01 [30.01, 30.02]	0.283	406.4	5T
80, 3	Triangle	0/5	30.02 [30.01, 30.02]	0.299	447.4	5T
80, 3	Risk, no root	0/5	30.05 [30.02, 30.07]	0.279	458.2	5T
80, 3	Global risk	0/5	30.00 [30.00, 30.01]	0.179	90.2	5T

Table 9: Symmetric mixture: primary gate comparison. Time is charged mean seconds [95% seed-bootstrap interval], with all outcomes retained. Gap is mean absolute upper-minus-lower bound; visits count distinct integer supports. O: numerical optimum; T: time limit; M: master failure; F: oracle failure; A: accuracy limit.

(D, d)	Method	Solved / n	Time (s)	Gap	Visits	Status
8, 2	Box	10/10	0.33 [0.31, 0.34]	1.0×10^{-7}	28.0	10O
8, 2	Triangle	10/10	0.32 [0.31, 0.33]	4.0×10^{-7}	27.9	10O
8, 2	Risk, no root	10/10	0.34 [0.33, 0.36]	6.0×10^{-7}	28.0	10O
8, 2	Global risk	10/10	0.68 [0.53, 0.87]	1.3×10^{-9}	26.7	10O
8, 2	Enumeration	10/10	0.17 [0.16, 0.17]	1.3×10^{-9}	28.0	10O
12, 3	Box	10/10	3.52 [3.40, 3.63]	1.3×10^{-9}	220.0	10O
12, 3	Triangle	10/10	4.75 [4.38, 5.08]	1.0×10^{-7}	219.2	10O
12, 3	Risk, no root	10/10	13.22 [12.30, 14.07]	1.0×10^{-7}	200.0	10O
12, 3	Global risk	10/10	16.39 [15.24, 17.52]	1.0×10^{-7}	197.0	10O
12, 3	Enumeration	10/10	1.37 [1.35, 1.39]	1.3×10^{-9}	220.0	10O
20, 3	Box	0/5	30.01 [30.00, 30.02]	0.263	804.0	5T
20, 3	Triangle	0/5	30.03 [30.01, 30.05]	0.256	806.0	5T
20, 3	Risk, no root	0/5	30.00 [30.00, 30.00]	0.236	301.8	5T
20, 3	Global risk	0/5	30.00 [30.00, 30.00]	0.151	188.8	5T
40, 3	Box	0/5	30.01 [30.00, 30.02]	0.263	655.8	5T
40, 3	Triangle	0/5	30.01 [30.00, 30.02]	0.283	692.6	5T
40, 3	Risk, no root	0/5	30.04 [30.02, 30.06]	0.268	582.2	5T

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(D, d)	Method	Solved / n	Time (s)	Gap	Visits	Status
40, 3	Global risk	0/5	30.00 [30.00, 30.00]	0.176	127.2	5T
80, 3	Box	0/5	30.01 [30.00, 30.01]	0.287	419.2	5T
80, 3	Triangle	0/5	30.01 [30.00, 30.03]	0.288	460.4	5T
80, 3	Risk, no root	0/5	30.02 [30.01, 30.04]	0.291	442.2	5T
80, 3	Global risk	0/5	30.00 [30.00, 30.01]	0.190	96.8	5T

On 159 solved gate/enumeration pairs, the maximum absolute upper-objective difference is 0.000. Interval incompatibilities at combined numerical tolerances: 0. Enumeration references are paired by mechanism, dimension, budget, radius, and seed. A comparison with no completed reference is not counted as exact objective agreement.

C.3 Testing, recovery and independent prediction

Table 10: Mean difference: common-support testing and recovery. Means [95% training-seed intervals]. All methods use budget three; the null level is 0.05.

Selector	n	Power	Null rejection	Recall
Global risk	30	0.928 [0.848, 0.978]	0.052 [0.035, 0.068]	0.533 [0.467, 0.589]
Robust greedy	30	0.965 [0.942, 0.987]	0.053 [0.038, 0.070]	0.589 [0.522, 0.656]
MMD screening	30	0.992 [0.980, 1.000]	0.050 [0.033, 0.068]	0.756 [0.689, 0.822]
Additive MMD	30	0.982 [0.962, 0.997]	0.050 [0.033, 0.065]	0.733 [0.656, 0.811]
L1 logistic	30	0.977 [0.950, 0.997]	0.043 [0.028, 0.060]	0.722 [0.656, 0.800]
Random	30	0.587 [0.435, 0.742]	0.048 [0.030, 0.067]	0.233 [0.156, 0.311]

Table 11: Mean difference: independently refitted common heads. Randomized and thresholded balanced error are distinct; AUC evaluates ranking. Values are means [95% seed intervals].

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓
Global risk	30	0.448 [0.437, 0.459]	0.382 [0.360, 0.404]	0.650 [0.625, 0.676]	0.315 [0.297, 0.336]
Robust greedy	30	0.447 [0.437, 0.457]	0.380 [0.359, 0.400]	0.657 [0.633, 0.681]	0.306 [0.288, 0.325]
MMD screening	30	0.433 [0.425, 0.442]	0.346 [0.329, 0.364]	0.694 [0.673, 0.714]	0.279 [0.265, 0.293]
Additive MMD	30	0.439 [0.429, 0.449]	0.360 [0.339, 0.383]	0.679 [0.654, 0.703]	0.286 [0.269, 0.306]
L1 logistic	30	0.436 [0.427, 0.446]	0.354 [0.334, 0.373]	0.688 [0.665, 0.710]	0.281 [0.266, 0.295]
Random	30	0.482 [0.474, 0.490]	0.456 [0.437, 0.474]	0.559 [0.536, 0.583]	0.405 [0.377, 0.435]

Table 12: Variance difference: common-support testing and recovery. Means [95% training-seed intervals]. All methods use budget three; the null level is 0.05.

Selector	n	Power	Null rejection	Recall
Global risk	30	0.360 [0.255, 0.475]	0.040 [0.025, 0.057]	0.433 [0.322, 0.544]
Robust greedy	30	0.292 [0.202, 0.383]	0.047 [0.028, 0.067]	0.411 [0.322, 0.500]
MMD screening	30	0.405 [0.302, 0.515]	0.048 [0.037, 0.060]	0.533 [0.433, 0.622]
Additive MMD	30	0.328 [0.222, 0.435]	0.050 [0.035, 0.067]	0.444 [0.344, 0.544]
L1 logistic	30	0.182 [0.115, 0.255]	0.045 [0.030, 0.060]	0.289 [0.222, 0.367]
Random	30	0.198 [0.142, 0.265]	0.050 [0.030, 0.073]	0.233 [0.156, 0.311]

Table 13: Variance difference: independently refitted common heads. Randomized and thresholded balanced error are distinct; AUC evaluates ranking. Values are means [95% seed intervals].

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓
Global risk	30	0.482 [0.476, 0.488]	0.486 [0.477, 0.496]	0.565 [0.543, 0.585]	0.420 [0.398, 0.443]
Robust greedy	30	0.479 [0.473, 0.484]	0.476 [0.467, 0.485]	0.574 [0.555, 0.594]	0.425 [0.404, 0.444]
MMD screening	30	0.472 [0.465, 0.478]	0.466 [0.454, 0.478]	0.595 [0.574, 0.616]	0.400 [0.378, 0.420]
Additive MMD	30	0.477 [0.470, 0.484]	0.476 [0.464, 0.489]	0.576 [0.554, 0.599]	0.418 [0.394, 0.442]
L1 logistic	30	0.484 [0.478, 0.489]	0.477 [0.468, 0.487]	0.555 [0.537, 0.573]	0.447 [0.430, 0.463]
Random	30	0.489 [0.484, 0.493]	0.487 [0.476, 0.496]	0.540 [0.525, 0.556]	0.454 [0.438, 0.470]

Table 14: Equal marginals, different dependence: common-support testing and recovery. Means [95% training-seed intervals]. All methods use budget three; the null level is 0.05.

Selector	n	Power	Null rejection	Recall
Global risk	30	0.050 [0.032, 0.068]	0.048 [0.033, 0.063]	0.256 [0.156, 0.356]
Robust greedy	30	0.027 [0.015, 0.040]	0.060 [0.045, 0.077]	0.222 [0.144, 0.311]
MMD screening	30	0.053 [0.035, 0.073]	0.052 [0.033, 0.073]	0.267 [0.178, 0.367]
Additive MMD	30	0.043 [0.025, 0.063]	0.047 [0.027, 0.072]	0.222 [0.144, 0.300]
L1 logistic	30	0.052 [0.033, 0.073]	0.050 [0.038, 0.062]	0.300 [0.211, 0.389]
Random	30	0.065 [0.045, 0.085]	0.047 [0.030, 0.063]	0.233 [0.156, 0.311]

Table 15: Equal marginals, different dependence: independently refitted common heads. Randomized and thresholded balanced error are distinct; AUC evaluates ranking. Values are means [95% seed intervals].

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓
Global risk	30	0.497 [0.494, 0.500]	0.492 [0.483, 0.499]	0.510 [0.500, 0.520]	0.494 [0.488, 0.500]
Robust greedy	30	0.499 [0.497, 0.501]	0.496 [0.490, 0.503]	0.504 [0.497, 0.512]	0.492 [0.486, 0.498]
MMD screening	30	0.497 [0.494, 0.500]	0.491 [0.481, 0.499]	0.509 [0.501, 0.521]	0.490 [0.482, 0.497]
Additive MMD	30	0.497 [0.495, 0.499]	0.494 [0.488, 0.500]	0.509 [0.502, 0.516]	0.493 [0.486, 0.500]
L1 logistic	30	0.498 [0.496, 0.500]	0.494 [0.486, 0.500]	0.507 [0.499, 0.515]	0.494 [0.489, 0.499]
Random	30	0.499 [0.497, 0.501]	0.496 [0.490, 0.502]	0.503 [0.497, 0.509]	0.497 [0.492, 0.502]

Table 16: Symmetric mixture: common-support testing and recovery. Means [95% training-seed intervals]. All methods use budget three; the null level is 0.05.

Selector	n	Power	Null rejection	Recall
Global risk	30	0.770 [0.657, 0.867]	0.055 [0.040, 0.070]	0.633 [0.567, 0.700]
Robust greedy	30	0.808 [0.705, 0.897]	0.055 [0.040, 0.073]	0.689 [0.611, 0.767]
MMD screening	30	0.808 [0.703, 0.902]	0.052 [0.035, 0.068]	0.711 [0.633, 0.789]
Additive MMD	30	0.763 [0.643, 0.872]	0.058 [0.042, 0.075]	0.656 [0.589, 0.722]
L1 logistic	30	0.215 [0.127, 0.313]	0.050 [0.033, 0.067]	0.233 [0.156, 0.311]
Random	30	0.260 [0.170, 0.362]	0.052 [0.033, 0.073]	0.233 [0.156, 0.311]

Table 17: Symmetric mixture: independently refitted common heads. Randomized and thresholded balanced error are distinct; AUC evaluates ranking. Values are means [95% seed intervals].

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓
Global risk	30	0.423 [0.408, 0.438]	0.350 [0.322, 0.381]	0.721 [0.686, 0.753]	0.314 [0.290, 0.340]
Robust greedy	30	0.411 [0.395, 0.427]	0.325 [0.294, 0.358]	0.746 [0.709, 0.780]	0.297 [0.269, 0.325]
MMD screening	30	0.405 [0.390, 0.420]	0.313 [0.285, 0.342]	0.754 [0.723, 0.785]	0.289 [0.262, 0.316]
Additive MMD	30	0.417 [0.403, 0.429]	0.332 [0.304, 0.358]	0.737 [0.707, 0.766]	0.314 [0.285, 0.342]
L1 logistic	30	0.479 [0.468, 0.488]	0.462 [0.438, 0.481]	0.568 [0.542, 0.599]	0.445 [0.424, 0.466]
Random	30	0.478 [0.467, 0.488]	0.458 [0.435, 0.477]	0.569 [0.542, 0.600]	0.439 [0.414, 0.462]

Table 18: Paired global-risk-minus-marginal-screening power. Negative values mean lower observed power for global robust-risk selection; the interval resamples training-seed differences.

Mechanism	Paired n	Δ power [95% CI]
Mean difference	30	-0.063 [-0.143, -0.013]
Variance difference	30	-0.045 [-0.197, 0.110]
Equal marginals, different dependence	30	-0.003 [-0.028, 0.020]
Symmetric mixture	30	-0.038 [-0.203, 0.117]

All six selectors above buy the same number of coordinates. Known-signal and all-feature diagnostics are reserved for the appendix. Within-seed two-sample batches were averaged before uncertainty was computed; conditional batches are not counted as independent selections. Valid predictive metrics from a time-limited or failed global search describe its retained support and do not certify optimal selection.

C.4 Real-data prediction and selection

Table 19: Wisconsin diagnostic breast cancer: mean ± split SD, not an independent-population confidence interval. Worst stress is the maximum over the three prescribed shifts. Effective metric counts and selection statuses are retained in the appendix.

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓	Stress BER ↓
Global risk	20	0.251±0.046	0.112±0.031	0.965±0.020	0.081±0.017	0.119±0.030
Robust greedy	20	0.220±0.032	0.099±0.021	0.973±0.013	0.080±0.021	0.106±0.020
MMD screening	20	0.179±0.013	0.091±0.021	0.973±0.011	0.084±0.017	0.098±0.021
Additive MMD	20	0.180±0.013	0.090±0.019	0.975±0.010	0.082±0.014	0.096±0.019
L1 logistic	20	0.230±0.034	0.097±0.035	0.971±0.017	0.078±0.020	0.106±0.035
Random	20	0.310±0.066	0.192±0.089	0.893±0.093	0.156±0.078	0.199±0.088

Table 20: Wisconsin diagnostic breast cancer: charged selection time and original-risk bounds. A dash means no global original-objective certificate, not zero risk. Risk statuses use O/T/M/F/A as in the computational tables.

Selector	n	Time (s) ↓	Mean LB	Mean UB	Status
Global risk	20	30.00±0.00	0.143	0.296	20T
Robust greedy	20	0.54±0.02	—	—	baseline
MMD screening	20	0.00±0.00	—	—	baseline
Additive MMD	20	0.02±0.00	—	—	baseline
L1 logistic	20	0.00±0.00	—	—	baseline
Random	20	0.00±0.00	—	—	baseline

Table 21: Ionosphere: mean ± split SD, not an independent-population confidence interval. Worst stress is the maximum over the three prescribed shifts. Effective metric counts and selection statuses are retained in the appendix.

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓	Stress BER ↓
Global risk	20	0.300±0.045	0.243±0.059	0.869±0.053	0.251±0.094	0.247±0.062

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Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓	Stress BER ↓
Robust greedy	20	0.285±0.035	0.217±0.052	0.885±0.045	0.202±0.081	0.223±0.052
MMD screening	20	0.284±0.044	0.223±0.055	0.865±0.055	0.195±0.053	0.229±0.053
Additive MMD	20	0.273±0.042	0.209±0.047	0.877±0.051	0.181±0.058	0.216±0.047
L1 logistic	20	0.284±0.034	0.227±0.050	0.880±0.046	0.198±0.049	0.232±0.047
Random	20	0.330±0.028	0.270±0.047	0.835±0.034	0.303±0.069	0.276±0.048

Table 22: Ionosphere: charged selection time and original-risk bounds. A dash means no global original-objective certificate, not zero risk. Risk statuses use O/T/M/F/A as in the computational tables.

Selector	n	Time (s) ↓	Mean LB	Mean UB	Status
Global risk	20	30.00±0.00	0.140	0.353	20T
Robust greedy	20	0.64±0.02	—	—	baseline
MMD screening	20	0.00±0.00	—	—	baseline
Additive MMD	20	0.02±0.00	—	—	baseline
L1 logistic	20	0.00±0.00	—	—	baseline
Random	20	0.00±0.00	—	—	baseline

C.5 Radius sensitivity

Table 23: Mean difference: radius sensitivity, means [95% training-seed intervals]. The primary radius remains 0.25. Supports are refitted at each radius; no result-selected setting is declared optimal.

Radius	n	Power ↑	Random BER ↓	Stress BER ↓	Status
0.10	10	0.790 [0.595, 0.915]	0.494 [0.491, 0.497]	0.488 [0.478, 0.498]	10O
0.25	10	0.855 [0.650, 0.980]	0.466 [0.450, 0.480]	0.441 [0.409, 0.468]	10O
0.50	10	0.955 [0.915, 0.990]	0.426 [0.411, 0.440]	0.380 [0.354, 0.404]	10O

Table 24: Symmetric mixture: radius sensitivity, means [95% training-seed intervals]. The primary radius remains 0.25. Supports are refitted at each radius; no result-selected setting is declared optimal.

Radius	n	Power ↑	Random BER ↓	Stress BER ↓	Status
0.10	10	0.495 [0.295, 0.710]	0.486 [0.478, 0.493]	0.499 [0.483, 0.514]	10O
0.25	10	0.630 [0.425, 0.845]	0.446 [0.417, 0.469]	0.402 [0.343, 0.450]	1M, 9O
0.50	10	0.970 [0.930, 0.995]	0.371 [0.352, 0.392]	0.256 [0.215, 0.300]	1M, 9O

C.6 Component time and effort

Table 25: computational, Mean difference: all-outcome mean component times, calls, root bounds, and feasible original risk. These means include unsuccessful outcomes; an absent component is a dash.

D	Method	n	Oracle (s)	Master (s)	Calls	Root LB	Risk UB
8	Box	10	0.17	0.13	29.1	0.111	0.506
8	Triangle	10	0.17	0.14	29.1	0.111	0.506
8	Risk, no root	10	0.17	0.19	28.0	0.000	0.506
8	Global risk	10	0.24	0.45	38.0	0.169	0.506
8	Enumeration	10	—	—	28.0	—	0.506
12	Box	10	1.54	2.51	243.4	0.063	0.273
12	Triangle	10	1.40	3.21	221.3	0.063	0.273
12	Risk, no root	10	1.36	12.35	203.2	0.000	0.273
12	Global risk	10	1.36	15.16	202.1	0.111	0.273
12	Enumeration	10	—	—	220.0	—	0.273
20	Box	5	7.10	22.85	809.4	0.038	0.314
20	Triangle	5	7.14	22.81	813.4	0.038	0.317
20	Risk, no root	5	2.74	27.24	306.4	0.000	0.311
20	Global risk	5	1.66	28.32	182.8	0.108	0.311
40	Box	5	7.20	22.77	654.0	0.015	0.302
40	Triangle	5	7.62	22.34	694.6	0.015	0.302
40	Risk, no root	5	6.30	23.69	584.0	0.000	0.292
40	Global risk	5	1.45	28.55	127.0	0.092	0.310
80	Box	5	6.22	23.76	407.4	0.007	0.290
80	Triangle	5	6.77	23.20	448.4	0.007	0.306
80	Risk, no root	5	6.86	23.15	458.2	0.000	0.279
80	Global risk	5	1.60	28.40	94.2	0.091	0.281

Table 26: computational, Symmetric mixture: all-outcome mean component times, calls, root bounds, and feasible original risk. These means include unsuccessful outcomes; an absent component is a dash.

D	Method	n	Oracle (s)	Master (s)	Calls	Root LB	Risk UB
8	Box	10	0.18	0.14	29.3	0.114	0.477
8	Triangle	10	0.17	0.14	29.3	0.114	0.477
8	Risk, no root	10	0.17	0.18	28.0	0.000	0.477
8	Global risk	10	0.23	0.44	37.0	0.176	0.477
8	Enumeration	10	—	—	28.0	—	0.477
12	Box	10	1.39	2.11	221.0	0.062	0.275
12	Triangle	10	1.40	3.34	220.2	0.062	0.275
12	Risk, no root	10	1.32	11.88	200.0	0.000	0.275
12	Global risk	10	1.35	15.02	201.2	0.111	0.275
12	Enumeration	10	—	—	220.0	—	0.275
20	Box	5	7.03	22.91	805.0	0.035	0.298
20	Triangle	5	7.04	22.93	807.0	0.035	0.291
20	Risk, no root	5	2.68	27.30	301.8	0.000	0.299
20	Global risk	5	1.74	28.24	192.8	0.106	0.295
40	Box	5	7.25	22.71	656.8	0.015	0.278
40	Triangle	5	7.57	22.39	693.6	0.015	0.298
40	Risk, no root	5	6.28	23.71	582.2	0.000	0.268
40	Global risk	5	1.48	28.51	131.4	0.093	0.287

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D	Method	n	Oracle (s)	Master (s)	Calls	Root LB	Risk UB
80	Box	5	6.30	23.67	420.2	0.007	0.293
80	Triangle	5	6.94	23.04	461.4	0.007	0.295
80	Risk, no root	5	6.51	23.48	442.2	0.000	0.291
80	Global risk	5	1.67	28.32	100.8	0.093	0.293

Table 27: ratio, Mean difference: all-outcome mean component times, calls, root bounds, and feasible original risk. These means include unsuccessful outcomes; an absent component is a dash.

D	Method	n	Oracle (s)	Master (s)	Calls	Root LB	Risk UB
8	Global risk	5	0.20	0.41	32.4	0.180	0.477
8	Complete ratios	5	0.20	0.60	32.0	0.180	0.477
12	Global risk	5	1.40	14.58	202.0	0.113	0.297
12	Complete ratios	5	1.20	24.25	165.8	0.113	0.297
20	Global risk	5	1.35	28.63	183.0	0.096	0.257
20	Complete ratios	5	0.71	29.28	88.6	0.096	0.256

Table 28: ratio, Symmetric mixture: all-outcome mean component times, calls, root bounds, and feasible original risk. These means include unsuccessful outcomes; an absent component is a dash.

D	Method	n	Oracle (s)	Master (s)	Calls	Root LB	Risk UB
8	Global risk	5	0.20	0.33	31.2	0.179	0.470
8	Complete ratios	5	0.20	0.64	31.0	0.179	0.470
12	Global risk	5	1.37	14.15	196.6	0.122	0.280
12	Complete ratios	5	1.15	24.94	162.4	0.122	0.280
20	Global risk	5	1.34	28.64	183.8	0.092	0.243
20	Complete ratios	5	0.67	29.32	87.4	0.092	0.244

C.7 Diagnostic supports and probability loss

Table 29: Reference supports are diagnostic, not six-method budget-matched competitors. Known signal uses ground truth; all features exceed the acquisition budget. Values are means [95% training-seed intervals].

Diagnostic reference	n	Power ↑	Null rejection	Random BER ↓
Mean difference		1.000	0.053	0.414
Known signal	30	[1.000, 1.000]	[0.038, 0.070]	[0.405, 0.422]
Mean difference		0.998	0.055	0.497
All features	30	[0.995, 1.000]	[0.037, 0.073]	[0.496, 0.497]
Variance difference		0.957	0.050	0.445
Known signal	30	[0.940, 0.973]	[0.035, 0.065]	[0.441, 0.450]
Variance difference		0.138	0.057	0.498
All features	30	[0.112, 0.167]	[0.038, 0.077]	[0.497, 0.498]
Equal marginals, different dependence		0.218	0.055	0.471
Known signal	30	[0.187, 0.250]	[0.040, 0.070]	[0.467, 0.474]
Equal marginals, different dependence		0.060	0.050	0.499
All features	30	[0.045, 0.077]	[0.033, 0.068]	[0.499, 0.499]
Symmetric mixture		1.000	0.048	0.351
Known signal	30	[1.000, 1.000]	[0.032, 0.065]	[0.342, 0.359]

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Diagnostic reference	n	Power ↑	Null rejection	Random BER ↓
Symmetric mixture		0.207	0.058	0.493
All features	30	[0.163, 0.253]	[0.038, 0.080]	[0.492, 0.494]

Table 30: statistical: predictive cross-entropy of the half-hinge-trained common head, semantic support stability, and charged selection time. Synthetic intervals are seed-bootstrap intervals; real entries use split SD. Stability is pairwise Jaccard, not a confidence interval.

Dataset / selector	Valid n	Log loss ↓	Stability ↑	Time (s) ↓
mean		0.670		21.15
Global risk	30	[0.653, 0.687]	0.246	[20.72, 21.56]
mean		0.666		0.21
Robust greedy	30	[0.649, 0.681]	0.276	[0.20, 0.21]
mean		0.642		0.00
MMD screening	30	[0.627, 0.657]	0.452	[0.00, 0.00]
mean		0.651		0.04
Additive MMD	30	[0.633, 0.670]	0.430	[0.04, 0.04]
mean		0.645		0.00
L1 logistic	30	[0.629, 0.661]	0.414	[0.00, 0.00]
mean		0.729		0.00
Random	30	[0.714, 0.745]	0.157	[0.00, 0.00]
variance		0.725		22.11
Global risk	30	[0.714, 0.737]	0.200	[21.16, 23.26]
variance		0.722		0.21
Robust greedy	30	[0.711, 0.733]	0.183	[0.21, 0.21]
variance		0.714		0.00
MMD screening	30	[0.703, 0.725]	0.248	[0.00, 0.00]
variance		0.723		0.04
Additive MMD	30	[0.712, 0.735]	0.209	[0.04, 0.04]
variance		0.737		0.00
L1 logistic	30	[0.727, 0.748]	0.179	[0.00, 0.00]
variance		0.744		0.00
Random	30	[0.734, 0.753]	0.157	[0.00, 0.00]
dependence		0.759		21.32
Global risk	30	[0.751, 0.766]	0.164	[20.94, 21.71]
dependence		0.762		0.21
Robust greedy	30	[0.755, 0.769]	0.164	[0.20, 0.21]
dependence		0.757		0.00
MMD screening	30	[0.750, 0.764]	0.169	[0.00, 0.00]
dependence		0.758		0.04
Additive MMD	30	[0.751, 0.764]	0.156	[0.04, 0.04]
dependence		0.761		0.00
L1 logistic	30	[0.753, 0.769]	0.167	[0.00, 0.00]
dependence		0.763		0.00
Random	30	[0.757, 0.770]	0.157	[0.00, 0.00]
mixture		0.614		22.24
Global risk	30	[0.590, 0.640]	0.316	[21.33, 23.31]
mixture		0.594		0.21
Robust greedy	30	[0.567, 0.623]	0.372	[0.21, 0.22]
mixture		0.588		0.00
MMD screening	30	[0.561, 0.614]	0.403	[0.00, 0.00]
mixture		0.606		0.04
Additive MMD	30	[0.581, 0.629]	0.338	[0.04, 0.04]
mixture		0.720		0.00
L1 logistic	30	[0.699, 0.738]	0.186	[0.00, 0.00]
mixture		0.723		0.00
Random	30	[0.702, 0.742]	0.157	[0.00, 0.00]

Table 31: real: predictive cross-entropy of the half-hinge-trained common head, semantic support stability, and charged selection time. Synthetic intervals are seed-bootstrap intervals; real entries use split SD. Stability is pairwise Jaccard, not a confidence interval.

Dataset / selector	Valid n	Log loss ↓	Stability ↑	Time (s) ↓
wdbc Global risk	20	0.329±0.059	0.179	30.00±0.00
wdbc Robust greedy	20	0.291±0.039	0.233	0.54±0.02
wdbc MMD screening	20	0.255±0.030	0.398	0.00±0.00
wdbc Additive MMD	20	0.252±0.029	0.387	0.02±0.00
wdbc L1 logistic	20	0.302±0.047	0.266	0.00±0.00
wdbc Random	20	0.431±0.105	0.062	0.00±0.00
ionosphere Global risk	20	0.480±0.079	0.082	30.00±0.00
ionosphere Robust greedy	20	0.513±0.232	0.153	0.64±0.02
ionosphere MMD screening	20	0.828±0.567	0.455	0.00±0.00
ionosphere Additive MMD	20	0.753±0.527	0.452	0.02±0.00
ionosphere L1 logistic	20	0.471±0.094	0.141	0.00±0.00
ionosphere Random	20	0.553±0.096	0.058	0.00±0.00

Table 32: Wang paper-objective weights with the independently frozen native additive statistic. This diagnostic evaluates the learned weighted statistic, rather than the common joint-RBF support test. Means [95% seed intervals].

Mechanism	Valid n	Native power ↑	Native null rejection
Mean difference	30	0.977 [0.950, 0.998]	0.045 [0.025, 0.065]
Variance difference	30	0.518 [0.408, 0.625]	0.052 [0.033, 0.072]
Equal marginals, different dependence	30	0.045 [0.030, 0.060]	0.052 [0.035, 0.070]
Symmetric mixture	30	0.945 [0.915, 0.970]	0.040 [0.025, 0.057]

C.8 Outcome inventory and effective counts

Table 33: Final outcomes aggregated by suite and method. O denotes a numerical optimum for the original robust criterion; T a selection limit; F a numerical failure. Other includes baseline completion and any remaining status. Complete counts refer to evaluation.

Suite / method	Total	Complete	O	T	F	Other
computational Box	70	70	40	30	0	0
computational Enumeration	40	40	40	0	0	0

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Suite / method	Total	Complete	O	T	F	Other
computational Global risk	70	70	39	30	1	0
computational Risk, no root	70	70	40	30	0	0
computational Triangle	70	70	40	30	0	0
ratio Global risk	30	30	20	10	0	0
ratio Complete ratios	30	30	17	13	0	0
real Robust greedy	40	40	0	0	0	40
real L1 logistic	40	40	0	0	0	40
real MMD screening	40	40	0	0	0	40
real Random	40	40	0	0	0	40
real Global risk	40	40	0	40	0	0
real Additive MMD	40	40	0	0	0	40
sensitivity Global risk	60	60	58	0	2	0
statistical All features	120	120	0	0	0	120
statistical Robust greedy	120	120	0	0	0	120
statistical L1 logistic	120	120	0	0	0	120
statistical MMD screening	120	120	0	0	0	120
statistical Random	120	120	0	0	0	120
statistical Global risk	120	120	114	6	0	0
statistical Known signal	120	120	0	0	0	120
statistical Additive MMD	120	120	0	0	0	120

Table 34: Effective counts aggregated for availability only. Test replicates are not extra trained selectors, and confidence intervals are computed within settings. Real-data testing power is undefined.

Suite / method	Total	Test n	Classifier n	Source check n
real Robust greedy	40	n/a	40	40
real L1 logistic	40	n/a	40	40
real MMD screening	40	n/a	40	40
real Random	40	n/a	40	40
real Global risk	40	n/a	40	40
real Additive MMD	40	n/a	40	40

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Suite / method	Total	Test n	Classifier n	Source check n
sensitivity				
Global risk statistical	60	60	60	60
All features statistical	120	120	120	120
Robust greedy statistical	120	120	120	120
L1 logistic statistical	120	120	120	120
MMD screening statistical	120	120	120	120
Random statistical	120	120	120	120
Global risk statistical	120	120	120	120
Known signal statistical	120	120	120	120
Additive MMD	120	120	120	120

There are 1260 explicit shifted-source certificate checks. The largest recorded nonnegative surrogate violation is 0.000; 0 exceed 10^{-7}. This directly budget-feasible source-atom calculation is separate from population stress.

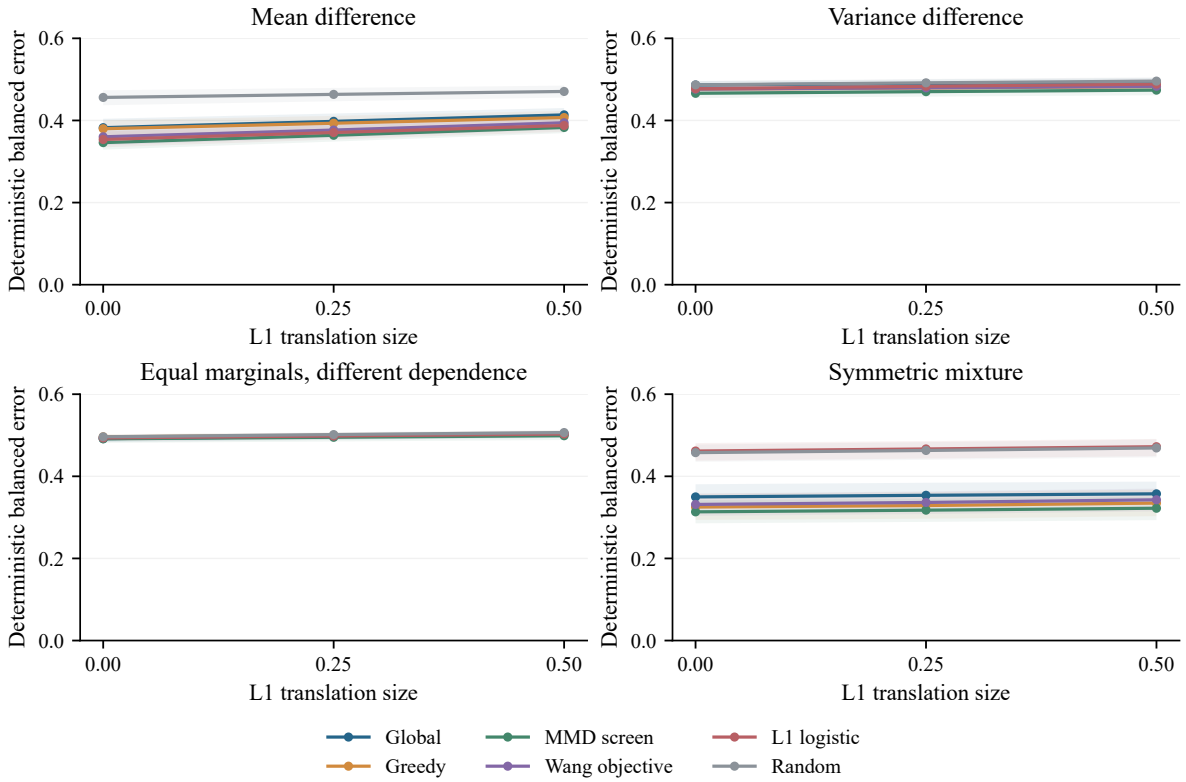


Figure 7: Independent prediction under the preset class-toward-class translations. The maximum over these shifts is distinct from the direct empirical-ball source check.

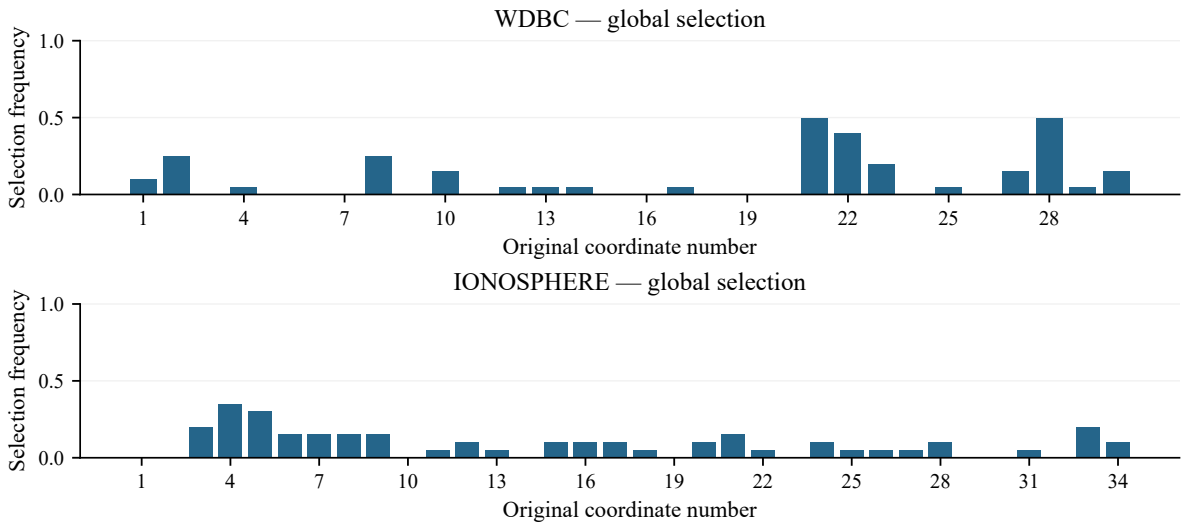


Figure 8: Global-selector feature frequencies over twenty repeated splits. These describe selection stability; the corresponding searches are time-limited incumbents.

D Exploratory full-fraction master

This supplement was declared after the primary timing observations and before its own runs. Objective, oracle, warm support, root stage and budget are fixed. A reusable repaired dual certificate with $\tau > 0$ gives

$$V(z) \geq f_c(z) = \frac{v}{\tau + \hat{\mu}^\top z}, \qquad 0 \leq v \leq \mathcal{H}(\hat{\Gamma}). \tag{6}$$

Dividing its masses and prices by the denominator restores normalization at any new support. For a feasible original-test bound U, shared products $y_j = \eta z_j$ have the exact binary representation

$$\begin{gathered} 0 \leq \eta \leq U, \quad 0 \leq y_j \leq U z_j, \quad y_j \leq \eta, \quad y_j \geq \eta - U(1 - z_j), \\ v \leq \tau\eta + \hat{\mu}^\top y, \qquad \mathbf{1}^\top y = d\eta. \end{gathered} \tag{7}$$

An original optimum with $\eta = \mathrm{OPT}$, $y = \mathrm{OPT}\, z$ remains feasible, preserving the global bound. The master retains affine floors as well: the product relaxation at fractional masks need not dominate a matched tangent, even though the full fraction does. The incumbent level row $\hat{\mu}^\top z \geq v/U - \tau$ is already implied by (7). Zero-scale and zero-numerator certificates are omitted; a feasible zero-risk test proves optimality by nonnegativity. The companion note derives these claims, actual-scale tangents and numerical margins in detail.

Mean difference — supplementary readout. At $D = 8$, the observed mean ratio-minus-tangent charged time is +0.19 s over 5 paired seeds; complete ratios solve 5/5 recorded instances and tangents 5/5. At $D = 12$, the observed mean ratio-minus-tangent charged time is +9.45 s over 5 paired seeds; complete ratios solve 3/5 recorded instances and tangents 5/5. At $D = 20$, the observed mean ratio-minus-tangent charged time is +0.00 s over 5 paired seeds; complete ratios solve 0/5 recorded instances and tangents 0/5. Negative time differences favor the supplement on these paired instances. The table gives seed-level uncertainty and achieved gaps; no universal speed or superiority claim follows.

Symmetric mixture — supplementary readout. At $D = 8$, the observed mean ratio-minus-tangent charged time is +0.31 s over 5 paired seeds; complete ratios solve 5/5 recorded instances and tangents 5/5. At $D = 12$, the observed mean ratio-minus-tangent charged time is +10.57 s over 5 paired seeds; complete ratios solve 4/5 recorded instances and tangents 5/5. At $D = 20$, the observed mean ratio-minus-tangent charged time is +0.00 s over 5 paired seeds; complete ratios solve 0/5 recorded instances

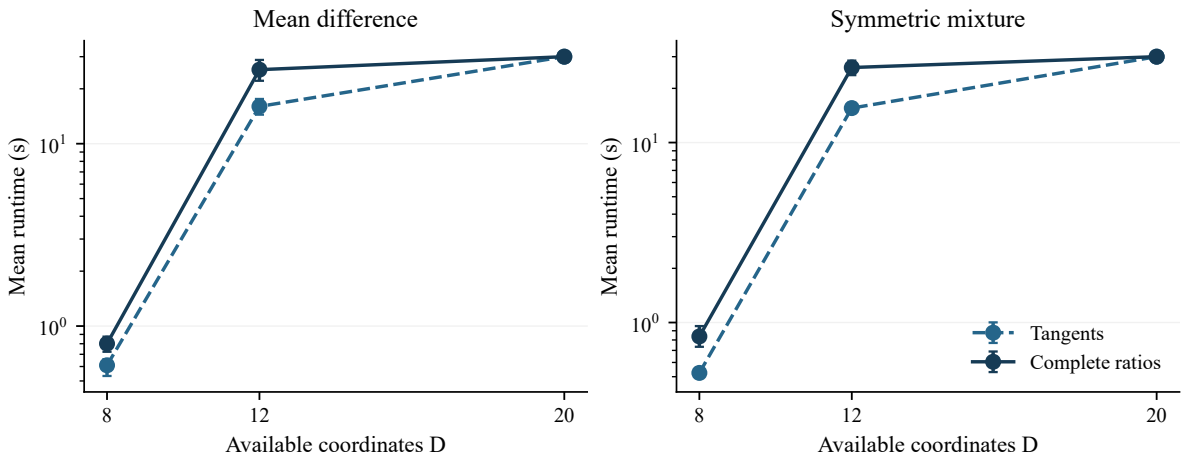


Figure 9: Separately declared paired full-fraction comparison. Seeds 10–14 are used within this suite; primary-suite timings are not substituted. Both variants retain failures and limits in their solved fraction.

and tangents 0/5. Negative time differences favor the supplement on these paired instances. The table gives seed-level uncertainty and achieved gaps; no universal speed or superiority claim follows.

Table 35: Mean difference: paired exploratory master comparison. Time is charged mean seconds [95% seed-bootstrap interval], with all outcomes retained. Gap is mean absolute upper-minus-lower bound; visits count distinct integer supports. O: numerical optimum; T: time limit; M: master failure; F: oracle failure; A: accuracy limit.

(D, d)	Method	Solved / n	Time (s)	Gap	Visits	Status
$8, 2$	Global risk	5/5	0.61 [0.53, 0.66]	2.0×10^{-7}	27.8	5O
$8, 2$	Complete ratios	5/5	0.80 [0.72, 0.88]	1.2×10^{-9}	27.4	5O
$12, 3$	Global risk	5/5	16.01 [14.43, 17.59]	2.0×10^{-7}	197.8	5O
$12, 3$	Complete ratios	3/5	25.46 [22.13, 28.79]	0.009	161.6	3O, 2T
$20, 3$	Global risk	0/5	30.00 [30.00, 30.00]	0.115	178.8	5T
$20, 3$	Complete ratios	0/5	30.00 [30.00, 30.00]	0.083	84.4	5T

Table 36: Within-seed ratio-minus-tangent charged time. Negative differences favor the supplement; these are descriptive paired intervals, not significance tests.

Dimension	Paired n	Δ time (s) [95% CI]
$D = 8$	5	0.19 [0.12, 0.25]
$D = 12$	5	9.45 [7.38, 11.39]
$D = 20$	5	0.00 [-0.00, 0.00]

Table 37: Symmetric mixture: paired exploratory master comparison. Time is charged mean seconds [95% seed-bootstrap interval], with all outcomes retained. Gap is mean absolute upper-minus-lower bound; visits count distinct integer supports. O: numerical optimum; T: time limit; M: master failure; F: oracle failure; A: accuracy limit.

(D, d)	Method	Solved / n	Time (s)	Gap	Visits	Status
$8, 2$	Global risk	5/5	0.52 [0.50, 0.56]	4.0×10^{-7}	27.2	5O
$8, 2$	Complete ratios	5/5	0.84 [0.73, 0.96]	1.3×10^{-9}	27.0	5O
$12, 3$	Global risk	5/5	15.54 [14.63, 16.59]	4.0×10^{-7}	192.0	5O
$12, 3$	Complete ratios	4/5	26.11 [23.69, 28.54]	0.002	157.8	4O, 1T
$20, 3$	Global risk	0/5	30.00 [30.00, 30.00]	0.101	179.6	5T
$20, 3$	Complete ratios	0/5	30.00 [30.00, 30.00]	0.072	83.2	5T

Table 38: Within-seed ratio-minus-tangent charged time. Negative differences favor the supplement; these are descriptive paired intervals, not significance tests.

Dimension	Paired n	Δ time (s) [95% CI]
$D = 8$	5	0.31 [0.22, 0.40]
$D = 12$	5	10.57 [7.36, 13.79]
$D = 20$	5	0.00 [0.00, 0.00]

These observations use seeds 10–14 in the separately frozen supplement. The tangent comparator is from this suite itself; primary-suite timings are not substituted. The component-time tables in the appendix identify whether a difference comes from master or oracle work.

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